# Units & Dimensions — Practice Worksheet

Class 11 Physics · 9Education

Name: ______________________________ Date: ______________ Section: ________

*Answer every question. Show your working wherever needed.*

## Section A — Fundamental & Derived Quantities, SI Units

**1.** How many fundamental (base) quantities are there in the SI system?

**2.** Write the SI units of length, mass and time.

**3.** Is mass a fundamental quantity or a derived quantity?

**4.** Is speed a fundamental quantity or a derived quantity?

**5.** State whether temperature is a scalar or a vector.

**6.** State whether force is a scalar or a vector.

**7.** State whether distance is a scalar or a vector.

**8.** Is electric current a scalar or a vector?

**9.** What is the SI unit of electric current?

**10.** What is the SI unit of temperature?

**11.** What is the SI unit of amount of substance?

**12.** What is the SI unit of luminous intensity?

**13.** Is the newton a fundamental unit or a derived unit?

**14.** Which is the SI base unit of mass — the gram or the kilogram?

## Section B — Prefixes & Unit Conversions

**15.** What power of ten does the prefix "kilo" represent?

**16.** What power of ten does the prefix "milli" represent?

**17.** What power of ten does the prefix "mega" represent?

**18.** Write the value of 1 ångström (Å) in metres.

**19.** Which is bigger: 1 micrometre or 1 nanometre?

**20.** Which is larger: 1 microsecond or 1 millisecond?

**21.** Convert 5 km into metres.

**22.** Convert 7 µm into metres.

**23.** Convert 1 metre into centimetres.

**24.** Convert 1 hour into seconds.

**25.** Convert 90 km/h into m/s.

**26.** Convert 72 km/h into m/s.

**27.** How many $cm^2$ are there in 1 $m^2$?

**28.** How many $mm^2$ are there in 1 $cm^2$?

**29.** How many $cm^3$ are there in 1 $m^3$?

**30.** Convert 2 $m^3$ into $cm^3$.

## Section C — Dimensional Formulae

**31.** What do the symbols [M], [L] and [T] stand for?

**32.** Write the dimensional formula of area.

**33.** Write the dimensional formula of volume.

**34.** Write the dimensional formula of speed.

**35.** Write the dimensional formula of acceleration.

**36.** Write the dimensional formula of force.

**37.** Write the dimensional formula of density.

**38.** What does it mean when a quantity has the dimensional formula $[M^0L^0T^0]$?

**39.** Write the dimensional formula of momentum.

**40.** Write the dimensional formula of work (or energy).

**41.** Write the dimensional formula of power.

**42.** Write the dimensional formula of pressure.

**43.** Write the dimensional formula of impulse.

**44.** Write the dimensional formula of frequency.

**45.** Name one quantity that has the same dimensional formula as work.

**46.** Name one quantity that has the same dimensional formula as momentum.

**47.** Write the dimensional formula of the gravitational constant G.

**48.** Write the dimensional formula of Planck’s constant h.

**49.** Write the dimensional formula of strain.

**50.** Is refractive index dimensionless? (Yes / No)

**51.** Is an angle (in radians) dimensionful or dimensionless?

**52.** What must be the dimensions of the angle $\theta$ inside $\sin\theta$?

**53.** Give one example of a dimensionless physical quantity.

## Section D — Principle of Homogeneity & Checking Equations

**54.** State the Principle of Homogeneity in your own words.

**55.** Is the equation $v = u + at$ dimensionally correct? (Yes / No)

**56.** Check dimensionally whether $s = ut + ½at^2$ is correct.

**57.** Check dimensionally whether $v = u + at^2$ is correct.

**58.** Can you add a quantity of dimension [L] to one of dimension [T]?

**59.** The equation $v = u + 2at$ is dimensionally correct. Is it physically correct?

**60.** If the terms of an equation do NOT match dimensionally, what can you conclude about the equation?

**61.** If the terms of an equation DO match dimensionally, is the equation definitely correct?

## Section E — Unit Conversion by Dimensional Analysis

**62.** Using dimensions, how many ergs are there in 1 joule?

**63.** How many dynes are there in 1 newton?

**64.** What does the relation $n_1u_1 = n_2u_2$ tell you?

**65.** When converting a quantity whose dimensional formula contains $L^2$, to what power must you raise the length ratio?

66. If you measure a quantity using a bigger unit, does the numerical value become bigger or smaller?

## Section F — Deriving Formulae & Limitations

67. The period of a simple pendulum is $T = k\sqrt{(l/g)}$. Does it depend on the mass of the bob?
68. Can dimensional analysis find the value of the constant k (such as $2\pi$) in a derived formula?
69. The speed of a wave on a string comes out as $v = k\sqrt{(T/\mu)}$. What is the actual value of k?
70. Two students derive a result as $Q = Av^2\rho$ and $Q = 2Av^2\rho$. Both are dimensionally correct. Can dimensional analysis tell which one is right?
71. Can dimensional analysis be used to derive the equation $s = ut + ½at^2$?
72. State any one limitation of dimensional analysis.
73. Can dimensional analysis distinguish between work and torque?
74. Can a formula containing sin, cos or log be derived using dimensional analysis?
75. A physical quantity depends on 4 different variables. Can dimensional analysis find its formula?

## Section G — Significant Figures

76. How many significant figures are there in 234?
77. How many significant figures are there in 0.005?
78. How many significant figures are there in 2.50?
79. How many significant figures are there in 205?
80. How many significant figures are there in 0.00420?
81. How many significant figures are there in $6.022 \times 10^{23}$?
82. How many significant figures are there in 100.0?
83. How many significant figures are there in the number 2500?
84. How many significant figures are there in 0.0030500?

## Section H — Rounding & Arithmetic with Significant Figures

85. Round 7.892 to two decimal places.
86. Round 4.745 to two decimal places.
87. Round 2.675 to two decimal places.
88. Add 12.11 + 1.4 and write the answer to the correct number of decimal places.
89. Multiply $5.7 \times 3.4$ and write the answer to the correct number of significant figures.
90. Multiply $2.0 \times 3.000$ and write the answer to the correct number of significant figures.

## Section I — Errors in Measurement & Error Propagation

91. Name the three types of error in measurement.
92. Taking more readings and averaging reduces which type of error?
93. Which type of error cannot be reduced by taking more readings?
94. A screw gauge has a fault that makes every reading 0.02 mm too high. Which type of error is this?
95. Why do we take the mean of several repeated readings?
96. Five readings of a length are 2.10, 2.12, 2.11, 2.13 and 2.14 cm. Find the mean.

**97.** For the readings in Q96, find the mean absolute error.

**98.** Using your answers to Q96 and Q97, write the length in the form (value ± error).

**99.** The mean of a set of readings is 4.52 cm. One of the readings is 4.54 cm. What is the absolute error of this reading?

**100.** Write the formula for relative error.

**101.** Write the formula for percentage error.

**102.** A diameter is measured as (4.52 ± 0.01) cm. Find its percentage error.

**103.** For $R = R_1 + R_2$ with $R_1 = (24 \pm 0.5)\ \Omega$ and $R_2 = (8 \pm 0.3)\ \Omega$, what is the absolute error in R?

**104.** For Z = A × B, the percentage errors in A and B are 5% and 10%. What is the percentage error in Z?

**105.** For X = A – B with A = (5.0 ± 0.2) and B = (3.0 ± 0.1), what is the absolute error in X?

**106.** State the power rule: if $Z = A^n$, how is the relative error in Z related to the relative error in A?

**107.** A quantity I has a percentage error of 2%. What is the percentage error in $I^2$?

**108.** A quantity r has a percentage error of 5%. What is the percentage error in $r^3$?

**109.** A student says the percentage error in $I^2$ is $(2\%)^2 = 0.04\%$. Is this correct? If not, give the right answer.

**110.** For the volume of a sphere $V = (4/3)\pi r^3$, the radius r has a 5% error. What is the percentage error in V?

**111.** For $P = I^2R$, the error in I is 2% and the error in R is 4%. What is the percentage error in P?

**112.** For $g = 4\pi^2 L/T^2$, the error in L is 0.5% and the error in T is 0.5%. What is the percentage error in g?

**113.** In the formula $Z = A / B^2$, does the power on B add to or subtract from the total percentage error?

**114.** For kinetic energy $KE = ½mv^2$, the error in m is 1% and the error in v is 3%. What is the percentage error in KE?

**115.** For density $\rho = 4M/(\pi d^2 h)$, the errors are: M → 0.5%, d → 1%, h → 0.25%. What is the percentage error in ρ?

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*End of worksheet. Units & Dimensions, Class 11 Physics.*